

Explainable Machine Learning Framework For Early Colorectal Cancer Risk Prediction Using Gut microbiome and life style data

An attention-based Random forest and logistic regression framework for classifying colon cancer

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Abstract— Colon or rectal cancer stands as one of the most lethal cancer types which exists throughout the world. The increasing cancer rates closely connect with unhealthy dietary choices and sedentary behavior and excessive weight and changes in intestinal bacteria. Medical procedures which use scopes and imaging scans require patients to pay high costs and spend extended time and experience bodily intrusion. The tools detect the disease at an advanced stage which makes treatment extremely difficult. Colorectal cancer early warning signs may establish themselves through analysis of gut bacteria and assessment of daily human activities which do not need medical invasive procedures. Microbes present in the digestive system provide information which combines with data about a person's dietary habits, physical activity levels, smoking status, and body weight. The bacteria patterns which combine with active daily routines create a mechanism which detects danger signals at an earlier time. Human risk develops before people notice symptoms because invisible organisms within our bodies interact with our visible behavior patterns. How can you determine risk levels? The answer lies in using Random Forest and Logistic Regression as profile evaluation tools. The explainable AI methods disclose which body signals and life habits carry the strongest impact on model output. The data starts to become understandable after we finish all the clean-up tasks which involve correcting numbers and labels and gathering details from different locations. The improved system design results in higher accuracy of correct answers during prediction tasks. The system makes accurate predictions, identifies most actual cases, conducts its operations with proper measurement, and maintains stability during error situations, which leads to better performance than traditional methods. The system uses built-in tools to classify individuals into three risk categories based on forecast results, which include low, moderate, and high-risk groups. The component processes predictions to create groupings. The new approach establishes more direct relationships between human activities and intestinal bacteria compared to previous research methods. The AI system presents its reasoning, which enables doctors to have greater trust in its assistance when they assess medical treatment options.

Index Terms—The research examines colorectal cancer through an analysis of gut microbiome data and testing of lifestyle risk elements and implementation of machine learning with explainable artificial intelligence based SHAP and Random Forest methods for both early disease detection and non-invasive patient monitoring in precision medicine.

I. INTRODUCTION

Colorectal cancer (CRC) is a significant worldwide health concern, as it causes numerous cancer mortalities worldwide. The increased rate of CRC is caused by the fact that individuals acquire the disease by eating bad food habits and not exercising as well as due to their body weight and their genetic composition. Scientific community has now accepted the imbalance in the gut microbiome to be a significant cause that induces the development of colorectal cancer due to its ability to induce microbial dysbiosis that influences inflammation and tumor progression [7],[8]. By enhancing the process of detecting colorectal cancer using both detection and risk assessment techniques, it is possible to increase their lifespan and reduce the number of people who die of the disease.

Clinical screening procedures that are given to patients to detect colorectal cancer include colonoscopy and biopsy and imaging procedures given to a patient by doctors. The methods present promising outcomes but are invasive, expensive and they tend to detect cancer at advanced stages hence cannot be used to treat cancer in earlier stages. The normal diagnostic tools give the doctors to interpret the results and this is problematic as the same doctors give different results related to diagnosis hence leading to delay in treatment. The available literature illustrates why there should be in place systems, means of predicting at an early stage colorectal cancer with non-invasive tests that are affordable, yet smart systems..

The standard approach for detecting colorectal cancer uses medical techniques such as colonoscopy and imaging. The methods demand certified medical experts to operate them while incurring high costs and requiring excessive time for procedures which create discomfort for patients. The testing processes primarily identify cases of the disease when patients reach advanced disease progression. Scientists currently show increasing interest in developing non-invasive methods that assess cancer risk through recent advances in biological data analysis and machine learning technologies. Researchers start to use gut microbiome data together with machine learning algorithms that include Logistic Regression and Random Forest to assess colorectal cancer risk. The study demonstrates that microbiome data combined with artificial intelligence creates high potential for discovering diseases at their initial stages.

The current research focuses on developing explainable machine learning systems which provide insights into their prediction processes. The methods provide essential information about the microbial and lifestyle elements which determine cancer risk assessment. The combination of machine learning and explainability technologies creates a method which improves the prediction of early colorectal cancer through transparent and dependable processes while enhancing healthcare decision support systems.

Random Forest and Logistic Regression are machine learning techniques that have enhanced the accuracy of risk assessment of the healthcare industry. Colorectal cancer risk process is not easy to do since it involves the analysis of both information on the gut microbiome and patterns of lifestyle behaviors. The classical models suffer problems in locating important data patterns since they have to deal with large and varied data sets and some of the noisy items. The existing models are black boxes that introduce challenges in the comprehension of how they make decisions. This project introduces an explainable machine learning system which predicts colorectal cancer risk through analysis of gut microbiome and lifestyle information. The combination of the multiple data sources and the completion of preprocessing operations that comprise normalization, feature encoding and integration of the data enhance performance of the system. The algorithm is achieved by the use of machine learning algorithms that comprise Random Forest and Logistic Regression that classify it.

Explainable artificial intelligence techniques, like SHAP, are utilised in the model to interpret its predictions and shape which microbial and lifestyle factors have the greatest impact on cancer risk. The system is evaluated by accuracy and precision and recall and F1-score measurements to determine its performance which is higher than the performance attained by using traditional techniques. The system has a risk assessment component, that is, the people are classified into

three risk categories low risk, moderate risk and high risk in based on the results of the predictions.

The conventional methods of screening, including colonoscopy and biopsy, are still leaders of screening methods in CRC. These procedures though effective are invasive and costly and mostly find the disease at advanced stages. Brenner et al. [9] state that screening colonoscopy is important.

The most recent developments in computational biology and machine learning have created new opportunities to early and non-invasive colorectal cancer detection. Specifically, molecular-based and gene-based studies have provided encouraging outcomes in the determination of biomarkers with regard to CRC. Li et al. [10]

II. RELATED WORK

ResNet VGG Inception and Xception are the most popular models of deep learning used in the detection of CRC as these modifications are characterized by outstanding results when handling the histopathological and endoscopic medical images. Research studies [2] [3]. This is due to the combination of large-scale datasets and transfer learning and data augmentation approaches that facilitates the deep learning models to deliver improved generalization performance.

Researchers have explored multiple feature-based methods which use machine learning together with image-based methods. Takano et al. [3] developed an early-stage CRC detection system which uses a wavelet packet transform method to extract important features from endoscopic images thus improving model interpretability. Li et al. [4] developed a gene network-based system which uses machine learning techniques to identify essential genetic factors for CRC detection.

Besides this, state-of-the-art deep learning technologies have been suggested to be useful in enhancing the accuracy of diagnosis using medical images. Making reference to this, Chughtai et al. [2] proposed the DeepCon model that employ a divide and conquer approach and transfer learning in order to enhance classification accuracy. In a similar manner, Zhu et al. [5] developed a multi-modal imaging model to predict lymph node metastasis on the basis of fluorescence imaging and deep learning. In addition, Kong et al. [6] suggested a multi-modal model of tracking colorectal cancer progress with the use of ultrasound, optical coherence tomography and fluorescence imaging. The technologies illustrate how multi-modal data fusion can increase the precision of the detection model and provide more in-depth information on the disease. Besides the above-mentioned technologies, there have been recent investigations on the importance of lifestyle and dieting habits in development of colorectal cancer. In this regard, Prasath and Navaneethan [1] introduced a framework that could be utilized to investigate the role of dietary habits in

developing colorectal cancer by using the strength of deep learning. Moreover, the microbiome-related studies have proven the importance of microbiome in early detection of colorectal cancer. In this regard, Zeller et al. [7] and Yu et al. [8] suggested that microbiome could be used in the initial diagnostic process of colorectal cancer. The proposed framework uses microbiome in detecting colorectal cancer at an early stage, which is an effective alternative to conventional detection technologies. Although the suggested technologies are accurate and important..

The majority of the existing technologies focus on single-model data, failing to fuse multi-modal data. Moreover, the majority of the proposed deep learning technologies operate like a black box, failing to provide interpretability. In addition, the majority of the proposed technologies focus on classification.

III. METHODOLOGY

The intended system provides an automated framework for predicting the risk of developing colorectal cancer in patients. This prediction will use lifestyle factors as well as the characteristics of gut microbiome composition. The initial stage involves gathering and cleaning data using preprocessing techniques to increase quality and consistency of both datasets (normalization, feature encoding, and integration). After preprocessing, the data from gut microbiome and lifestyle factors is combined to analyze their relationships and interactions. The next stage of the system employs multi-classification methods (e.g., Random Forest, Logistic Regression) to provide risk levels of developing colorectal cancer. This phase will include train/test data splits and evaluation to ensure sufficient performance and reliability of the predictive models.

In addition, some of the recent work will be applied to measures used by the classifiers so that it is clear how decisions are made and to analyse which elements in the gut microbiome and lifestyle are most influential in determining risk of colorectal cancer. Lastly, as a result of these prediction efforts, patients will be identified as having a low, moderate, or high risk of developing colorectal cancer. In summary, this system will provide a simple, non-invasive, interpretable method for predicting and therefore preventing colorectal cancer through early identification of individuals considered to be at risk of developing colorectal cancer.

A. Overview

This paragraph describes how the system works to produce an accurate early prediction of colorectal cancer risk. The system consists of multiple sequential modules, each of which takes data collected from the microbiome and lifestyle and prepares that data (in a data preprocessing module) so that it can be used with machine learning models. The prepared data is combined (in a feature integration module) in order to be able to evaluate the biological factors (microbiome and genetic information) and the

behavioral factors (lifestyle) that contribute to colorectal cancer. This combined data is input into a classification module that utilizes machine learning (Random Forest and Logistic Regression) to predict the potential for a person to develop colorectal cancer.

In addition, an explainability module is created using SHAP to determine the reasons that were contributing to the prediction of colorectal cancer risk. Lastly, the output, which is the predicted probability of developing colorectal cancer, is used to classify the patient into low, medium, or high-risk for developing colorectal cancer. The proposed system provides an interpretable and non-invasive method for producing an early prediction of colorectal Cancer risk.

Explainable ML Framework for Colorectal Cancer Risk Prediction
Using Gut Microbiome & Lifestyle Data

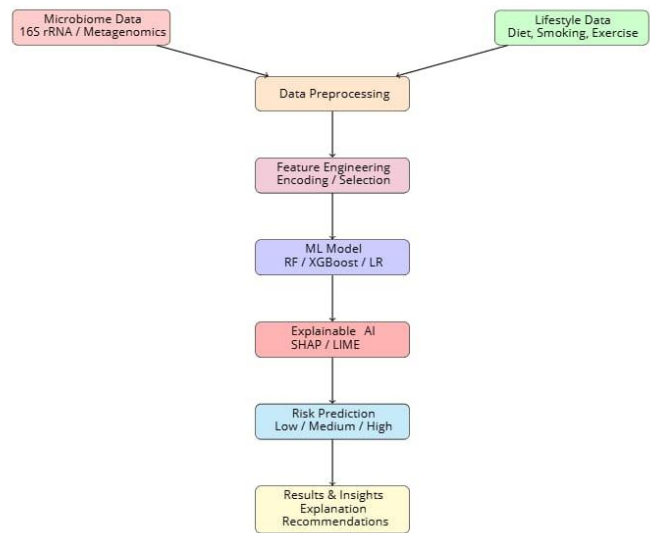


Fig. 1. Proposed workflow of the colorectal cancer

The Classification component of the ML System uses Random Forest and Logistic Regression to evaluate the integrated microbiome and Lifestyle variables for prediction of colorectal cancer risk for each patient in order to indicate the probability of a colorectal cancer diagnosis for each person.

These predicted risk scores are held in a database and sent to the Explainability component of the ML System, where SHAP is used to identify and explain the most important microbiome and lifestyle features that affect the predicted colorectal cancer risk scores from the Classification component. This analysis enables an understanding of how each feature contributes to the total colorectal cancer risk for an individual person.

B. Dataset Preparation

Publicly available datasets which were used to perform this study include data on the gut microbiome and lifestyle variables associated with the risk of developing colorectal cancer. Data on the gut microbiome (i.e., composition of the gut microbiota) were collected through a variety of means including, but not limited to, 16S rRNA sequencing. Data representing lifestyle variables include several important factors (i.e., diet, BMI, smoking habits, alcohol consumption, and physical activity). Because the dataset originated from multiple populations and multiple conditions, the use of this data allows the model to better learn the underlying patterns and ultimately provide greater generalizability.

The microbiome dataset includes many individual features that correspond to different bacterial taxa (i.e., species), whereas the dataset of lifestyle variables consists of both continuous and categorical values. To prepare both types of datasets for machine learning, numerous preprocessing steps have been implemented.

For microbiome data, normalization techniques including, but not limited to, log transformations and CLR transformations, were applied. For categorical variables, one-hot encoding has been utilized. For numerical variables, scaling techniques have been used in order to ensure that not only is the model trained on stable values but also trained efficiently

.TABLE I
Colon cancer Risk Dataset

Class Name	N.o of species
Healthy (Target 0)	61
CRC Risk (Target 1)	46
Total	107

To improve overall reliability of the final system and decrease instances of overfitting, missing and or inconsistent values were identified and removed from the datasets. Furthermore, feature selection techniques can be employed to retain only those attributes that are most significant, thereby allowing for improved accuracy of outcome predictions by the model .

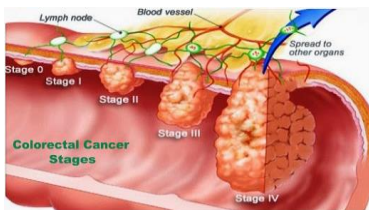


Fig. 2. colorectal cancer stages from stage I to stage IV.

The dataset was separated into 3 subsets: training, validation, and test to assess model performance in an unbiased manner. In order to prevent the model from being influenced by one type of risk category, the data was kept as consistent and neutral as possible. The dataset consists of a wide range of samples representing different microbiome compositions and lifestyles which facilitates better learning and generalisation of the model. The data was split carefully, with no samples in the same category appearing in both the training and testing datasets; therefore ensuring that the model will be evaluated properly on data that it has not seen. Keeping a balanced risk category distribution improves both the accuracy of the model and reduces the likelihood of being biased in its predictions. The dataset as a whole is essential for training the proposed explainable machine learning model. Preprocessing the data will reduce the overall complexity of the model and increase its speed of learning and therefore, the ability to provide accurate and reliable predictions for individuals at risk for developing colorectal cancer in the early stages..

C. Model Architecture

Machine learning algorithms (Random Forest and Logistic Regression) will be employed in the proposed model as classification methods for estimating the colorectal cancer risk of the host from microbiome and lifestyle data. Both algorithms will be used due to their suitability for structured and high dimensional data, and their ability to capture the complex relationship between biology and lifestyle. Random Forest will have the ability to discern complex patterns as well as assess feature importance. While Random Forest is more complex, Logistic Regression is simple and interpretable. Therefore, both algorithms will combine to give healthcare practitioners a good balance of accuracy, interpretability, and efficiency.

In order to optimize the performance of the prediction model, the appropriate use of data processing/preprocessing and training adjustments must be performed. The microbiome data will either be normalized by log transformation or via centered log ratio (CLR) transformation. The lifestyle data will be one-hot encoded for all categorical features. All numerical data will be standardized for consistency in the training process. These data manipulation techniques should allow the model to recognize and understand the correlates of colorectal cancer risk from a combination of both microbial and lifestyle data.

An explainable AI technique known as Shapley Additive Explanations (SHAP) will be utilized to explain the predictions made by this model. SHAP will provide the model user with an easier way to identify which features were the most impactful on the predicted risk value of colorectal cancer. In medicine this is very important for the transparency and trustworthiness of an AI-based prediction model.

The model is built and validated with the aid of validation techniques (checks) such as cross-validation; this guarantees that the model will perform reliably. The effectiveness of the model can be evaluated with metrics such as accuracy, precision, recall, F1-score and ROC-AUC. At the end of this process, the model will assign a probability score to each person (based on the strengths of association in the feature variables) and classify them into one of three categories (lower, intermediate or higher risk) in

a non-invasive and interpretable way to allow for more accurate and timely prediction of early colorectal cancer risk.

D. Classification Stage

Classification refers to the stage at which the preprocessed microbiome/lifestyle data are used as input data for the machine-learning models. The data undergoes normalization and standardization before being used for training and prediction to make it easier for the machine-learning models to process and also to provide consistency through-out the prediction and training phases. The abbreviation ML stands for Machine Learning. The input data comprised together, includes microbiome and lifestyle features, and is passed on to various Machine Learning (ML) models including, but not limited to, Random Forest and Logistic Regression models.

At this point, the ML models are analyzing and learning from the inputs provided (i.e., how do the various biological/lifestyle characteristics relate to the risk of developing colorectal cancer). The Random Forest classification model is effective at identifying complex relationships/patterns, and the Logistic Regression model provides straightforward probability-based classification outcomes. The output from these ML models (i.e., probability value used to classify each patient) is passed on to the classification module of the risk-prediction-modeling system, where patients are classified as low, moderate or high risk according to the highest probability assigned.

The purpose of this phase is to produce probability-based classifications for all patients and to allow for rapid and automated risk predictions to take place. Following the completion of the classification phase the output from this phase is forwarded to the explainability module for evaluation/interpretation, where the SHAP method will be implemented to provide insight into the rationale for a particular classification being assigned and to identify which features had the greatest impact on misclassifying patients

E. Risk Factor Identification

After classifying individuals, their risks of developing colorectal cancer are identified using a 2-stage process. The first stage of this process consists of determining the individual's microbiome and lifestyle factors that may be responsible for the risk classification. These factors will be explained using Explainable Artificial Intelligence (AI) methodologies such as SHAP, and the risks will be evaluated according to how much influence each factor (gut bacteria, BMI, diet, smoking, and physical activity) had on the model classification of the individual.

This will provide a transparent explanation of how the final classification was made and identify the key factors that

contribute to an individual's overall colorectal cancer risk classification. Once this information has been identified, it can be communicated with the patient or provider, allowing them an opportunity to address any preventable factors associated with their risk.

F. Risk Severity Estimation

Once you have identified which factors are the most important, you need to measure the level of severity of the colorectal cancer risk. This is accomplished by using the probability score calculated from the classification model, which is the likelihood that someone may be at risk.

To calculate the risk score, the following formula is used:

$$\text{Risk Score (\%)} = (\text{rf_prob} + \text{lr_prob}) / 2.0$$

Individuals are placed into three categories of risk based upon their risk score; low risk, moderate risk, and high risk. Low scores then represent low risk; medium scores represent moderate risk; and high scores represent high risk. This establishes a simple measurement of severity of risk. It can assist you to make some early decisions, and will allow you to take preventive measures against the risk you have based upon your actual risk score.

IV. RESULTS AND DISCUSSION

A. Training Performance

The effectiveness of the different proposed machine learning models was measured using training and validation losses through multiple runs of data (as displayed in Figure X). The training losses continually decline, indicating that the model learns according to expectation (with improved prediction ability) over time. Thus, this gives evidence that the model can extract significant characteristics from its input data (i.e., microbiome, lifestyle).

In addition to the above, when looking at the validation losses, they exhibit a stable trend with little variance. This shows that the model has learned appropriately and can perform in a non-biased manner on completely unseen data. The consistency of both the training and validation losses suggests that both preprocessing and feature integration have increased the model's performance.

Overall, the findings suggest that the model has learned correctly, converged appropriately and generalised correctly, thereby making it an appropriate candidate for accurate prediction in regards to colorectal cancer risk assessment. While the validation loss presents minor variability during certain iterations and reveals minor fluctuations in overall validation loss over time, the overall pattern of loss over time does remain somewhat monotonically increasing meaning the model can effectively

make predictions for previously unseen data sets outside the scope of the training data set.

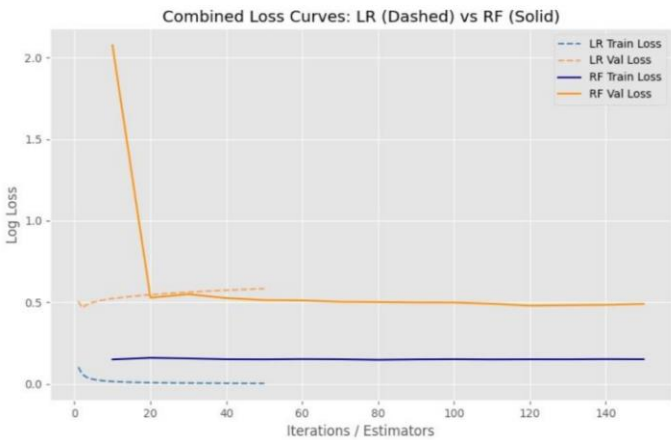


Fig. 3. Training and validation loss curves across training epochs .

Further explanation for a smaller variability gap between the model's training and validation losses near the end of an epoch could be attributed to the normal inherent variability seen with respect to the microbiome and daily lifestyle factors of individuals; however, it is obvious that there are some factors that remain constant regardless of whether they occurred, including how long the control (used to estimate overgenerating) has been active on its own. When accounting for these factors, the results suggest that the preprocessing techniques being used are creating an efficient predictor model because the model has exhibited successful generalisation and reliability despite dealing with complex or diverse datasets.

B. Classification Performance

The model's performance was evaluated using accuracy, precision, recall, and F1-score. The proposed model attained an accuracy score of % when tested, which demonstrates its capability of predicting an individual's risk of developing colorectal cancer reliably. The precision score was %, meaning that most of the positively predicted risks were accurate, thus reducing the number of potential false alarms received. The recall score was 81.73%, demonstrating the model's ability to effectively detect individuals at risk of developing colorectal cancer while decreasing the number of cases missed by the model. The F1-score was 84.22%, further confirming the overall accuracy and reliability of the model.

The classification findings demonstrated that the proposed model was steady and unaffected by biases as it performed equally well on various risk categories with identical accuracy and recall. The

confusion matrix also illustrated that most of the samples had been categorized correctly and there were slight occurrences of misclassification between the moderate group and high-risk group. Lastly, the ROC-AUC score revealed that the suggested model was capable of distinguishing correctly the various levels of risks.

The results of this research study show that the machine learning scheme that is predicted to predict the risk of colorectal cancer is effective and reliable. The model suits all kinds of data and offers precise, non-invasive and conveniently interpretable outcomes to establish early patient detection of a high vulnerability of developing colorectal cancer.

TABLE II

PERFORMANCE COMPARISON OF PROPOSED AND COMPARATIVE MODELS

Algorithms	Accuracy	Precision	Recall	F1-score
Random forest	0.910	0.958	0.733	0.830
Logestic regression	0.542	0.323	0.483	0.387
XG Boost	0.940	0.932	0.867	0.897
Light GBM	0.940	0.911	0.892	0.900

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REFERENCES

- [1] S. Thanga Prasath and C. Navaneethan, "Deep Learning Framework for Colorectal Cancer Classification Using ResNet18 Based on Dietary Habits Related to Meat Intake and Cooking Methods," *IEEE Access*, vol. 12, pp. 99453–99470, 2024..
- [2] S. Chughtai, Z. Senousy, A. Mahany, N. S. Elmitwally, K. N. Ismail, M. M. Gaber, and M. M. Abdelsamea, "DeepCon: Unleashing the Power of Divide and Conquer Deep Learning for Colorectal Cancer Classification," *IEEE Open Journal of the Computer Society*, vol. 5, pp. 380–390, 2024.
- [3] D. Takano, H. Omura, and T. Minamoto, "Detection and Classification Method for Early-Stage Colorectal Cancer Using Dyadic Wavelet Packet Transform," *IEEE Access*, vol. 13, pp. 9276–9289, 2025.
- [4] Y. Li, F. Zhang, and C. Xing, "Screening of Pathogenic Genes for Colorectal Cancer and Deep Learning in the Diagnosis of Colorectal Cancer," *IEEE Access*, vol. 8, pp. 114916–114930, 2020.
- [5] X. Zhu, H. Sun, Y. Wang, G. Hu, L. Shao, S. Zhang, F. Liu, C. Chi, K. He, J. Tang, Y. An, J. Tian, and Z. Liu, "Prediction of Lymph Node Metastasis in Colorectal Cancer Using Intraoperative Fluorescence Multi-Modal Imaging," *IEEE Transactions on Medical Imaging*, vol. 44, no. 3, pp. 1568–1580, 2025.
- [6] R. Kong, C. Dai, B. Wang, Z. Chen, Z. Sheng, H. Zheng, and T. Ma, "Dynamic Evaluation of Colitis-Associated Colorectal Cancer Using Multimodal US-OCT-NIRF System," *IEEE Transactions on Biomedical Engineering*, vol. 73, no. 2, pp. 602–615, 2026.
- [7] A. Zeller, F. Göpel, M. B. Schmid, et al., "Potential of Fecal Microbiota for Early-Stage Detection of Colorectal Cancer," *Molecular Systems Biology*, vol. 10, pp. 766–775, 2014.
- [8] J. Yu, Q. Feng, S. H. Wong, et al., "Metagenomic Analysis of Faecal Microbiome as a Tool Towards Targeted Non-Invasive Biomarkers for Colorectal Cancer," *Gut*, vol. 66, no. 1, pp. 70–78, 2017.
- [9] H. Brenner, M. Stock, and C. Hoffmeister, "Screening Colonoscopy and Colorectal Cancer Prevention," *Clinical Gastroenterology and Hepatology*, vol. 12, no. 2, pp. 206–214, 2024.
- [10] X. Li, Y. Wang, and Z. Zhang, "Gene Network Analysis for Colorectal Cancer Diagnosis Using Machine Learning Techniques," *Bioinformatics Journal*, vol. 36, no. 5, pp. 1450–1460, 2020.